

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)**

Term-End Examination

June, 2025

MCS-231 : MOBILE COMPUTING

Time : 3 Hours

Maximum Marks : 100

Note : *Question No. 1 is compulsory and carries 40 marks. Attempt any **three** questions from the rest.*

1. (a) Discuss the design considerations for mobile computing with a focus on operating system considerations and their impact. 10
- (b) Define WDM. Describe its advantages, disadvantages along with applications. 10
- (c) What is Wireless Application Protocol (WAP) ? How does it work ? Discuss WAP Protocol stack. 10

- (d) With reference to Smart Client Architecture, explain the following approaches : 10
- (i) Thick Client Architecture
 - (ii) Thin Client Architecture
2. (a) Describe the working of three-way handshake in TCP-protocol and give its key features. 10
- (b) Draw the diagram to depict the process of End-to-End connection in Mobile TCP. Give advantages of Mobile TCP. 10
3. (a) List and explain mobile device data store methods, briefly. 10
- (b) Define virtual network. Explain the types of virtual network and give its advantages. 10
4. (a) Explain three-tier architecture for mobile computing. 10
- (b) Explain the types of synchronization methods between mobile client and remote server. 10

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5. (a) Compare and contrast the features and advantages of FDM and TDM in the context to mobile communication. 10
- (b) Elaborate the method used for dynamic grouping of mobile devices to improve performance and to optimize the energy consumption. 10

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1. (a) What do you understand by mobility in mobile computing ? Explain different features of mobile computing. 10
 - (b) Explain GPRS along with its features. Describe the services offered by GPRS. 10

- (c) “Mobile computing often involves the use of TCP and UDP.” Compare and contrast these two protocols. 10
- (d) Write a short note about cells used in GSM. Explain the features of GSM. 10
2. (a) Draw a neat diagram for mobile computing architecture. Briefly explain it. 10
- (b) Elaborate the process of handing over control of a mobile device to a neighbouring cell. Discuss Soft Handover and Hard Handover. 10
3. (a) Explain the features of 5G Network. Describe its advantages. 10
- (b) Explain Mobile Device Data Store methods. 10

4. (a) Explain the objective of mobile IP. Explain the working of mobile IP with the help of a diagram. 10
- (b) Describe the concept of mobile cloud computing and its advantages for users. 10
5. (a) Explain data replication methods. 10
- (b) Explain thick and thin client architectures. 10

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December, 2022

MCS-231 : MOBILE COMPUTING

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Note : *Question No. 1 is compulsory and carries 40 marks. Attempt any **three** questions from the rest.*

1. (a) What is Frequency Division Multiplexing (FDM) ? What are its advantages, disadvantages and applications ? 10
- (b) What is GPRS ? Explain its features. Also, list the services offered by it. 10
- (c) What is LTE ? Explain its features. 10
- (d) Write a short note on the Adhoc Networks.

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2. (a) What is a cellular network ? Explain its various components. 10
(b) Explain mobile computing architecture with the help of a diagram. 10
3. (a) List any *five* applications of mobile computing. 10
(b) Explain J2EE architecture. 10
4. (a) Explain the functions of a mobile operating system. 10
(b) Explain the features of an integrated development platform for mobile apps. 10
5. (a) Explain the features of iOS. 10
(b) Explain the features of XML. 10

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1. (a) What is Time Division Multiplexing (TDM) ? What are the advantages, disadvantages and applications ? 10
 - (b) What is GSM technology ? Explain its advantages and disadvantages. 10
 - (c) What is a 4G network ? Explain its features. 10
 - (d) What is WIMAX ? Explain its features. 10
 2. (a) What is Data Synchronization ? Explain it. 10
 - (b) Write a short note on Mobile Commerce. 10

3. (a) Draw and explain Android architecture. 10
(b) Explain various components of J2ME. 10
4. (a) What is guided transmission ? How does it differ from unguided transmission ? Give example(s). 10
(b) Explain any *two* reasons for weakening of figural strength after transmission. 10
5. (a) What is an antenna ? What is its functionality ? On what factor(s) does its size depend on ? 10
(b) What is CDMA ? How does it differ from GSM ? 10

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1. (a) What is Time Division Multiplexing (TDM)? What are its advantages, disadvantages and applications ? 10
- (b) What is a Cellular Network ? Under what conditions frequency reuse is possible by different cells in the network ? 10
- (c) What are Smart Sensors ? What are the instances where mobile devices interact with their surroundings with the help of sensors ? 10
- (d) Write a short note on Mobile Device Database Management. 10

2. (a) List various steps in the development of a Mobile App. What are the various components of an Enterprise Mobile Application ? 10
(b) Write a short note on XML. 10
3. (a) What is J2ME ? Explain main components of J2ME. 10
(b) Write a short note on Android. 10
4. (a) Explain the features of integrated development platforms that are used for Mobile App development. 10
(b) Explain any *five* features of Mobile Communications. 10
5. (a) Explain the design considerations for Mobile Computing. 10
(b) Write a short note on Smart Appliances. 10

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1. (a) What is Wavelength Division Multiplexing ?
What are its advantages, disadvantages and applications ? 10
- (b) Explain WLAN Wi-Fi IEEE 802.11x networks. 10
- (c) What are Actuators ? Give some examples of Actuators found in Robots. 10

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- (d) Explain the architecture patterns for mobile device data store methods. 10
2. (a) Explain the architecture of a content management application. 10
- (b) What do you mean by XML parsing ? Explain with an example. 10
3. (a) Explain the architecture of J2EE. 10
- (b) Write a short note on SWIFT. 10
4. (a) Explain the architecture of iOS. 10
- (b) Explain various phases of development process of a mobile application. 10
5. (a) What is GPRS ? Explain its features. 10
- (b) What are adhoc networks ? Explain their advantages. 10